

Diagnosing Evidence Gaps in Hydrogen Energy Futures: A Semantic Retrieval Validation Framework

Input Question

What is the future of hydrogen energy?

Domain

Energy Science

Validation Status

needs_data

Problem Statement

The question seeks to understand the projected trajectories, technological bottlenecks, and economic viability of hydrogen energy systems. However, the currently available evidence base consists exclusively of literature on high-energy particle physics (colliders, Standard Model tests) and general Earth system detectability, which are topically irrelevant to hydrogen energy infrastructure, production, or consumption.

Rationale

Given the complete absence of domain-relevant evidence in the allowed set, no factual claims about hydrogen energy can be established. The primary scientific task is therefore to diagnose whether this absence is due to a systematic retrieval failure (semantic mismatch) or a genuine gap in the indexed corpus. This plan proposes a verifiable hypothesis testing framework to validate the retrieval system's performance using negative controls and targeted query expansion, rather than fabricating answers from unrelated physics data.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Semantic Retrieval Failure Hypothesis: The current evidence base lacks hydrogen energy content due to a systematic retrieval failure (semantic mismatch) rather than true absence of literature, verifiable via targeted re-retrieval with domain-specific expansion terms.
- **Mechanism**: The vector space distance between the query 'future of hydrogen energy' and the retrieved high-energy physics/earth science documents exceeds the relevance threshold. This gap is hypothesized to be an artifact of indexing or embedding limitations, not a reflection of the scientific corpus. Validating this requires demonstrating that specific hydrogen-related keywords yield high-relevance results that were previously missed.
- **Falsifiable Prediction**: Executing a re-retrieval campaign using the strictly logged query set ['hydrogen economy', 'green hydrogen production', 'PEM electrolysis efficiency', 'H2 storage materials', 'hydrogen fuel cell durability'] will retrieve at least one EvidenceCard with `relevance_score > 0.8`. If zero such cards are found after 3

iterations, the hypothesis of retrieval failure is weakened in favor of index coverage gaps.

- **Required Observations**: Execution of 3 retrieval iterations with the exact specified query expansion terms ; Logging of all query strings used to prevent post-hoc rationalization ; Relevance scores of all newly retrieved documents ; Count of documents exceeding relevance threshold 0.8
- **Risk of Being Wrong**: Moderate. Relevant literature may exist but remain unindexed or inaccessible even with expanded queries, leading to a false confirmation of retrieval failure when the issue is actually database scope.

Hypothesis 2

- **Hypothesis**: Negative Control Semantic Distance Hypothesis: The allowed evidence IDs (EV-Q116-*) are definitively irrelevant to hydrogen energy futures, as quantifiable semantic distance metrics will confirm their exclusion is scientifically justified and not arbitrary.
- **Mechanism**: Topical irrelevance is operationalized as a measurable semantic distance between document embeddings and the hydrogen energy query vector. By computing cosine similarity or equivalent metrics for each allowed ID against the query, we establish a quantitative baseline for exclusion. This prevents future attempts to force-fit unrelated physics/earth science content into hydrogen energy analysis.
- **Falsifiable Prediction**: All four allowed evidence IDs will exhibit semantic similarity scores < 0.3 against the hydrogen energy query embedding. If any ID shows similarity > 0.5 , the exclusion criterion is invalidated and potential cross-domain relevance must be re-evaluated.
- **Required Observations**: Computation of semantic distance metrics (e.g., cosine similarity) for each EV-Q116-* ID against 'future of hydrogen energy' query ; Documentation of the embedding model and version used for distance calculation ; Threshold validation confirming all scores fall below relevance cutoff
- **Risk of Being Wrong**: Low. Semantic distance metrics are well-established; primary risk is model bias toward certain domains, but this is mitigated by using multiple embedding models if initial results are borderline.

Hypothesis 3

- **Hypothesis**: Cross-Domain Methodology Transfer Null Hypothesis: Feasibility assessment frameworks from particle collider studies (EV-Q116-df35f895433236f4ceb28c22) cannot be validly applied to hydrogen energy futures without explicit bridging evidence, constituting a category error.
- **Mechanism**: While both domains involve 'future' planning, the underlying physical constraints (quantum field theory vs. electrochemical thermodynamics), economic scales, and stakeholder ecosystems are disjoint. Absent mediating evidence showing methodology transfer, applying collider feasibility structures to hydrogen roadmaps is invalid. This hypothesis serves as a guardrail against speculative analogical reasoning.
- **Falsifiable Prediction**: Citation network analysis of next 50 retrieved hydrogen energy documents will reveal zero citations to CERN FCC feasibility frameworks (EV-Q116-df35f895433236f4ceb28c22). Discovery of ≥ 1 such citation would invalidate the null hypothesis and suggest legitimate cross-domain methodology transfer.
- **Required Observations**: Citation network analysis linking EV-Q116-df35f895433236f4ceb28c22 to hydrogen policy/technical documents ; Semantic similarity scoring between collider feasibility sections and hydrogen roadmap sections ; Expert validation of methodological compatibility (if citations found)
- **Risk of Being Wrong**: High. Interdisciplinary methodology transfer does occur (e.g., systems engineering, project management frameworks), and dismissing it entirely may miss valid analogical reasoning pathways.

However, without direct evidence, asserting transferability remains speculative.

Technical Details

This experiment validates the 'Semantic Retrieval Failure Hypothesis' by quantifying the topical irrelevance of currently allowed evidence IDs (EV-Q116-*) and testing if targeted query expansion retrieves relevant hydrogen energy literature. The technical core involves two parallel tracks: (1) Negative Control Verification, which computes cosine similarity between embeddings of the allowed EV-Q116-* documents and a standard 'future of hydrogen energy' query vector to establish a baseline for exclusion; and (2) Active Retrieval Campaign, which executes three iterations of search using strictly logged, domain-specific query terms. The system uses a pre-trained sentence-transformer model (e.g., all-MiniLM-L6-v2 or similar) for embedding generation. Relevance is determined by a threshold of 0.8 on the cosine similarity score. All query strings and resulting document IDs are logged to prevent post-hoc rationalization.

Datasets

Source

```
```json
```

```
[
```

```
"Internal Literature Index containing EV-Q116-df35f895433236f4ceb28c22,
EV-Q116-31ebaf07631cc6937983e244, EV-Q116-1aa0198c2026b4b3143f4f51,
EV-Q116-2a7742bf1b9ed7bfd3373248",
```

```
"Query Expansion Set: ['hydrogen economy', 'green hydrogen production', 'PEM electrolysis efficiency', 'H2
storage materials', 'hydrogen fuel cell durability']"
```

```
]
```

```
```
```

Target

Relevance-scored EvidenceCards for Hydrogen Energy Domain and Semantic Distance Metrics for EV-Q116-* IDs

Paper Abstract

Background: The question of hydrogen energy's future requires synthesizing data on production, storage, and market dynamics. However, initial evidence retrieval yielded only high-energy physics and Earth system documents, indicating a potential semantic mismatch. Methods: We propose a two-stage validation protocol. First, we quantify the semantic distance (cosine similarity) between the query 'future of hydrogen energy' and the allowed irrelevant evidence IDs to establish a negative control baseline. Second, we execute a strictly logged re-retrieval campaign using domain-specific expansion terms to test for retrieval failure versus index coverage gaps. Validation Plan: Success is defined by retrieving at least one document with `relevance_score > 0.8` using expanded queries. Results: Pending execution of the verification experiments. No factual claims about hydrogen technology are made based on the current irrelevant evidence set.

Methods

1. **Negative Control Verification**: Generate embeddings for the text content of each allowed EV-Q116-* ID and the query 'future of hydrogen energy'. Compute pairwise cosine similarities. Document these scores to justify exclusion.
2. **Strictly Logged Re-retrieval**: Execute 3 retrieval iterations. In each iteration, use one or a combination of the predefined query expansion terms. Log the exact query string used for each iteration in an immutable log file.
3. **Relevance Scoring**: For all newly retrieved documents, compute relevance scores against the hydrogen energy query.
4. **Threshold Analysis**: Count documents with `relevance_score > 0.8`.
5. **Falsification Check**: If ≥ 1 document with score > 0.8 is found, the retrieval failure hypothesis is supported. If 0 such documents are found after 3 iterations, the hypothesis is weakened.

Experiments

Baselines

```
"""json
[
  "Baseline 1: Current State - Zero relevant hydrogen cards among allowed EV-Q116-* IDs.",
  "Baseline 2: Generic Query - Retrieval results using only 'energy future' without hydrogen-specific terms."
]
"""
```

Metrics

```
"""json
[
  "Cosine Similarity Score between EV-Q116-* embeddings and hydrogen query embedding",
  "Count of newly retrieved EvidenceCards with relevance_score > 0.8",
  "Precision@10 for the expanded hydrogen-specific queries"
]
"""
```

Ablation

Compare retrieval performance when using single-term queries (e.g., 'hydrogen economy') versus multi-term combined queries (e.g., 'green hydrogen production AND PEM electrolysis') to determine if specificity improves

relevance scoring.

Validation Protocol

1. Verify that all EV-Q116-* IDs have cosine similarity < 0.3 with the hydrogen query. 2. Review the top 5 retrieved documents from each re-retrieval iteration manually to confirm they discuss hydrogen energy technologies or policy. 3. Ensure the query log matches the executed searches exactly.

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- **EV-Q116-df35f895433236f4ceb28c22** · arxiv · arXiv:2511.20417
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2511.20417.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:7|section:page-7|paragraph:1;
content_sha256=7ce37b3278fcff7b1cb1e551eabb708c5e21e6f1ab0282b9dabf9f8bf528758
- **EV-Q116-31ebaf07631cc6937983e244** · arxiv · arXiv:0910.4753
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/0910.4753.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:2|section:page-2|paragraph:1;
content_sha256=9a69ba49d4dfedf130bb5df3b0b9ee2780d7cdd82ffc67d21642e698aff85453
- **EV-Q116-1aa0198c2026b4b3143f4f51** · arxiv · arXiv:hep-ph/9411291
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/hep-ph/9411291.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:5|section:page-5|paragraph:1;
content_sha256=23af71e76e678acc66db6dbf7a871dde265591ae261f98d9e322b48bbe1e97fe
- **EV-Q116-2a7742bf1b9ed7bfd3373248** · arxiv · arXiv:1902.08035
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1902.08035.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=df5243b4eb1392b71a96c7e7a93a4c61cb866a4d16936f6955d9cf6fec15ac5d

Reviewer Comments

- The revision successfully addresses previous required revisions by explicitly incorporating semantic distance metrics into the Negative Control Verification method and mandating immutable logging of query expansion terms.
- The 'Semantic Retrieval Failure Hypothesis' is correctly framed as a testable system-level hypothesis rather than an attempt to answer the hydrogen energy question with irrelevant physics data.
- Evidence grounding is strictly maintained; supporting_evidence_ids are used only to define the negative control set (irrelevant documents), avoiding any false attribution of hydrogen-related content to high-energy physics papers.
- Results field correctly remains 'pending' with no fabrication of experimental outcomes.
- Reproducibility is significantly improved through specific technical details (embedding model examples, threshold definitions, log immutability requirements).

Revision History

- auto_revision_1: 依据评审意见重做假设与实验设计。

Reproducibility Checklist

- Log exact query strings for all 3 retrieval iterations before execution.
- Record the specific embedding model version and parameters used for semantic distance calculation.
- Save the raw cosine similarity scores for each EV-Q116-* ID against the hydrogen query.
- Archive the list of all newly retrieved document IDs and their relevance scores per iteration.
- Provide the script used for computing cosine similarity and relevance scoring.